

Genome-wide quantification of TF binding at single DNA molecule resolution

Rozemarijn Kleinendorst^{1#}, Guido Barzaghi^{1,2#}, Mike L Smith¹, Judith B Zaugg³, Arnaud R Krebs^{1*}

¹European Molecular Biology Laboratory (EMBL), Genome Biology unit, Meyerhofstraße 1, 69117 Heidelberg. ²Faculty of Biosciences, Collaboration for Joint PhD Degree between EMBL and Heidelberg University, Heidelberg, Germany. ³European Molecular Biology Laboratory (EMBL), Structural and Computational Biology Unit, Meyerhofstraße 1, 69117 Heidelberg.

#These authors contributed equally to this work

*Correspondence should be addressed to: arnaud.krebs@embl.de

ABSTRACT

Precise control of gene expression requires the coordinated action of multiple factors at cis-regulatory elements (CREs). We recently developed Single Molecule Footprinting (SMF) to simultaneously resolve the occupancy of multiple proteins including Transcription Factors (TFs), RNA Pol II (Pol II) and nucleosomes on single DNA molecules genome-wide. The technique combines the use of cytosine methyltransferases to footprint the genome with bisulfite sequencing to resolve TF binding patterns at CREs. DNA footprinting is performed by incubating permeabilized nuclei with recombinant methyltransferases. Upon DNA extraction, whole genome or targeted bisulfite libraries are prepared and loaded on Illumina sequencers. The protocol can be completed in 4-5 days. Analysis can be performed in 2 days using a dedicated R package. The protocol can be executed in any laboratory with access to high-throughput sequencing. Our method can be used to analyse how TFs cooperate and antagonize to regulate transcription.

Keywords:

chromatin accessibility, DNA footprinting, Transcription Factor, gene regulation, RNA Pol II, DNA methylation.

INTRODUCTION

Transcription factors (TFs) modulate transcription through the recruitment of coactivators and RNA Polymerase II (Pol II) at the promoters of genes. There are several technologies available to directly measure the binding of transcriptional regulators (i.e. ChIP-seq¹, CUT&RUN²) or to indirectly infer protein occupancy through footprints in chromatin accessibility (DNase-seq³, ATAC-seq⁴). These methods have led to extensive insights on the identity of TFs involved in CRE activation in various cell types and tissues. Most TFs are unable to bind and activate their target CRE alone. Cooperativity between TFs has been shown to be an essential mechanism used by TFs to bind and activate cis-regulatory elements (CREs)⁵⁻⁸. Most genomics methods used to measure TF binding are bulk assays that typically average binding information from millions of cells. Most of these enrich for a single feature of interest (e.g. TF, chromatin mark...), disregarding the potential co-occurrence of other binding events and ignoring potential heterogeneity of occupancy at CREs.

Recently several approaches have been developed that employ exogenous DNA methyl-transferases to footprint protein-DNA contacts in the genome⁹⁻¹¹. These approaches were shown to accurately quantify DNA occupancy by nucleosomes⁹, TFs⁸, General Transcription Factors (GTFs) and Pol II¹⁰. Coupling methylation footprinting with various sequencing technologies has made it possible to resolve protein-DNA contacts continuously over several hundreds^{8,10} to several thousand¹²⁻¹⁵ base pairs on individual DNA molecules. This unprecedented resolution has enabled new insights into transcription initiation dynamics¹⁰, TF cooperativity⁸, transcriptional coordination¹³ and chromatin fiber organization^{12,14}.

Here we describe Single Molecule Footprinting (SMF), which we recently developed to resolve the occupancy of multiple TFs and Pol II simultaneously on single DNA molecules^{8,10}. The technique combines the use of cytosine methyltransferases to footprint the genome with bisulfite sequencing to resolve the molecular binding patterns of TFs at CREs. Footprinting is performed on permeabilized nuclei. It relies on the use of recombinant methyltransferases that are commercially available. Bisulfite libraries (and optional DNA capture step) can be performed using commercial protocols and are sequenced on Illumina MiSeq or NextSeq sequencers. The protocol can be executed in any molecular biology laboratory with access to high throughput sequencing and completed in 4-5 days. Subsequent data analysis can be performed in 2-3 days.

Continuity in footprinting information allows studying whether binding events occur simultaneously at CREs with molecular resolution. Specifically, it enables to quantify the degree of co-occupancy of TFs on the same DNA molecules and link their binding in a way that is impossible with bulk data. We have successfully used this strategy to study co-occupancy of TFs, to identify dependencies between TFs and to reveal cooperativity mechanisms underlying their action at CREs⁸.

OVERVIEW OF THE PROTOCOL

Experimental overview

SMF requires the extraction and permeabilization nuclei from cell lines or tissues. Purified nuclei are sequentially incubated with recombinant methyltransferases that methylate GpCs (McvPI) and/or CpGs (SssI) (Figure 1a). To obtain reproducible methylation footprints, it is essential to carefully quantify the number of cells used to maintain a constant enzyme/DNA substrate ratio. The number of cells to be used has to be adjusted per species according to its genome size (i.e. $2.5 \cdot 10^6$ for *Drosophila* and $0.25 \cdot 10^6$ for mouse or human). DNA is extracted, sheared in large molecules (300-500 bp) and bisulfite converted for whole genome DNA methylation profiling. When whole genome profiling is not suited, several targeted SMF strategies can be implemented (see sequencing strategy and coverage requirements). For studying TF binding in the mouse genome, we added a DNA capture-step using a library of RNA baits tiling 297,000 CREs. This step enriches libraries for regions of interest prior to bisulfite conversion and reduces the sequencing effort to 2% of the genome. With this strategy, a molecular coverage compatible with single molecule analysis ($>40\times$) can be reached at a large majority of TF binding sites with a reasonable sequencing effort ($200 \cdot 10^6$ reads) (Figure 1b,c). An alternative strategy consists of designing primers against regions of interest to prepare amplicon libraries that lead to very high molecular coverage ($>1000\times$) at defined regions with limited sequencing effort ($1 \cdot 10^6$ reads) (Figure 1d,e).

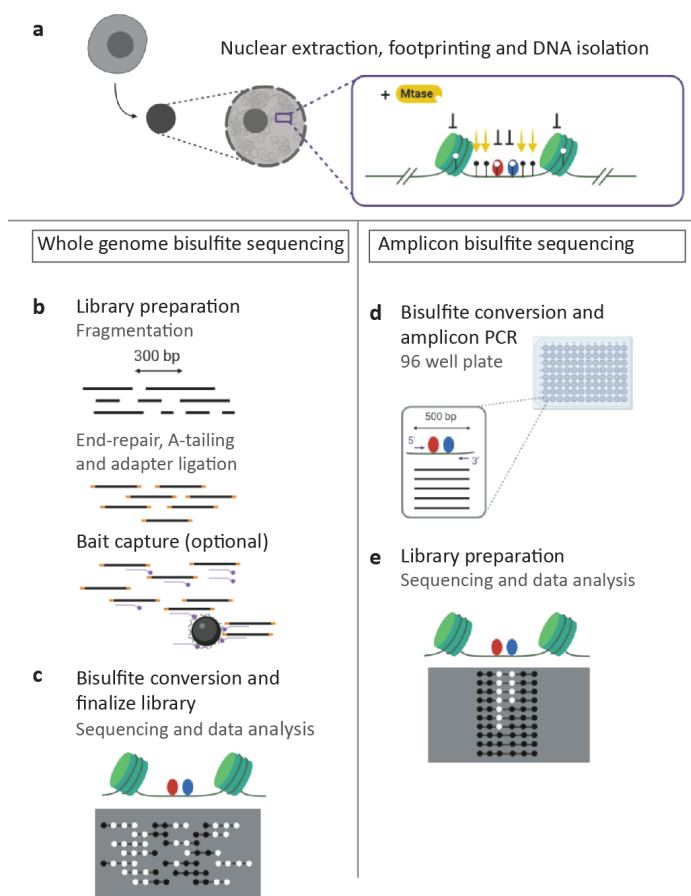


Figure 1: Overview of the experimental workflow. **a**, Nuclei are extracted using a hypotonic buffer. Methylation footprinting is performed by incubating the nuclei with a GpC (McvIP1), and optionally CpG (SssI) methyltransferase. Regions accessible to the enzymes are methylated, while regions bound by proteins (TFs, nucleosomes) are protected, creating footprints of various sizes. DNA is extracted and used for whole genome (left panel), or targeted amplicon analysis (right panel). For whole genome analysis, **b**, DNA is fragmented to a target size range of 300-500 bp. DNA is end-repaired and sequencing adapters are ligated. An optional capture step can be performed to enrich the library for regions of interest such as CREs and reduce the sequencing required for single molecule analysis. **c**, DNA is bisulfite converted and the library is amplified before sequencing on Illumina MiSeq and NextSeq platforms. **d**, Alternatively to the whole genome approach, primers can be designed to target 96 loci using amplicon bisulfite PCR. Amplicons are typically designed to cover 300-500 bp of the CRE. **e**, Amplicons are pooled, and library is prepared. Up to 12 libraries can be

multiplexed and sequenced on a MiSeq instrument. The read ends in amplicon data are identical for every molecule, creating focused high coverage views of the targeted loci.

Experimental Design

Footprinting efficiency

Preparation of the biological material for footprinting is key to successful SMF experiments. The number of cells to use has to be adjusted based on the genome size. The presented conditions allow efficient footprinting of $\sim 1 \mu\text{g}$ of DNA which corresponds to $0.25 \cdot 10^6$ mammalian cells or $2.5 \cdot 10^6$ Drosophila cells. This material is sufficient to prepare targeted (96 bis-PCR reactions) or whole genome bisulfite libraries. Homogeneity in nuclear extraction and permeabilization is important as the cytoplasmic membrane would prevent the penetration of the methyltransferases. This would lead to artefactual heterogeneity in the footprinting patterns (fully inaccessible molecules). It is therefore important to use a nuclear extraction protocol adapted to the cell type or tissue used. The current protocol is robust and has successfully been used for various fly cell lines (Schneider S2, OSC), and mammalian cell types (mESC, Neuronal Progenitors, MELs, C2C12, HeLa). It is however advisable to routinely check the homogeneity of the nuclear preparations before performing SMF using trypan blue.

Enzyme selection

SMF can be performed using the GpC methyltransferase McviPI alone or in combination with the CpG methyltransferase Sssl. The tandem treatment increases the spatial resolution of the assay. However, this can only be performed in cell types or tissue that do not have endogenous methylation signals in CpG context. We have successfully used that strategy in fly cell lines¹⁰ as well as mouse embryonic stem cells where endogenous methyltransferases are genetically depleted¹⁰. This is however not applicable to somatic cell types that do not survive depletion of endogenous methylation. Using only GpC methylation reduces the number of analysable binding sites by a factor of about two⁸. This nevertheless leaves several thousands of binding events representing each TF and is still useful to derive general rules about their function (see **Anticipated results**, Figure 7).

Sequencing strategy and coverage requirements

Sequencing a sufficient number of DNA molecules to cover the loci of interests is essential to enable accurate SMF analysis. The typical coverage reached in genome wide experiments is around 40 molecules. The coverage requirement varies as a function of the binding frequency of the studied protein. For TFs, we typically observed binding frequencies around 1-40%. While frequencies >20% would be accurately quantified with a coverage of 40x (8/40 molecules), lower binding frequencies would require higher sequencing depth. This consideration is even more critical when jointly analysing multiple binding events to allow accurate quantification of all the combinations of binding states. Calculations of the theoretical coverage should be conducted to decide on the sequencing strategy applied to footprinted DNA. For instance, performing a whole genome SMF sequencing experiment on a NextSeq 550 lane with 150 paired-end reads leads to $\sim 350 \cdot 10^6$ clusters of 300 bp (cost ~ 4500 EUR). Accounting for the lower mapping rates of bisulfite libraries ($\sim 60\%$), this achieves a theoretical coverage of 252x for the fly genome and of 20x for the mouse genome. In the second case, this is insufficient for single molecule analysis. In this case, targeted sequencing approaches should be considered. We advise to first perform a shallow sequencing run to evaluate the quality of the sequencing library ($\sim 1 \cdot 10^6$ reads) before generating a deeply sequenced dataset ($> 200 \cdot 10^6$ reads).

Quality controls

Several controls can be implemented to ensure the quality of SMF libraries. Sequencing accessibility over long DNA molecules (> 200 bp) is critical for the interpretation of SMF data at the single molecule level. Bisulfite conversion leads to significant DNA fragmentation. It is therefore important to experimentally determine the DNA fragment size distribution using Agilent Bioanalyzer (Figure 3a). Before investing in deep sequencing of the sample, it is also advised to generate low coverage data ($< 1 \cdot 10^6$ reads) in order to verify basic features of the libraries. These include the efficiency of bisulfite conversion, mapping rates,

complexity of the library (duplication rates), fragment size distribution (Figure 4b). Bisulfite conversion is estimated by calculating the average conversion of cytosines that are neither in CpG nor in GpC context. As these cytosines are not methylated *in vivo*, thymine frequency is expected to exceed 95% in this context. Additionally, capture efficiency can be estimated by calculating the fraction of mapped reads falling within the bait regions in the case of targeted enrichment in mammalian genomes.

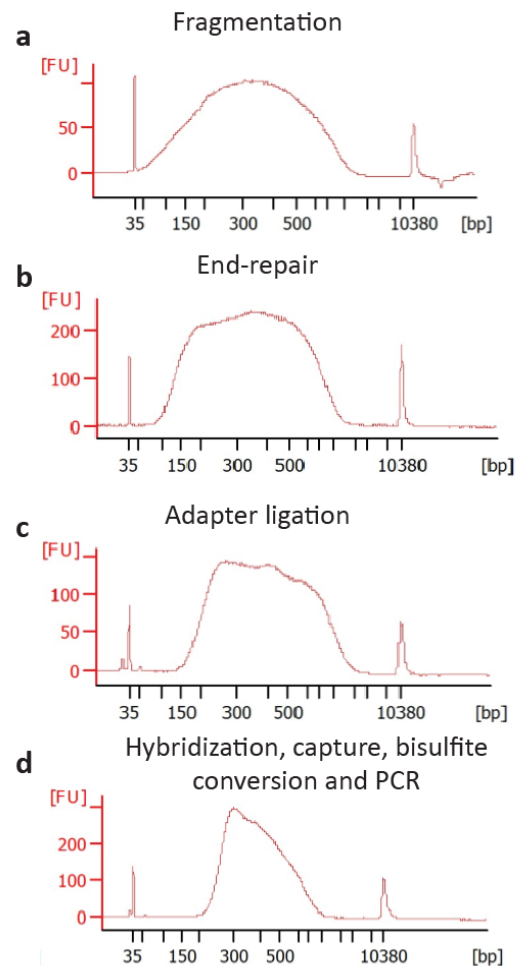


Figure 2: Quality controls during the preparation of the bait-captured SMF samples. Bioanalyzer traces after various steps of the protocol (step indicated). **a**, Footprinted DNA is fragmented with Covaris (300-500 bp). **b**, and subjected to end-repair and A-tailing. **c**, A ~50 bp shift in size distribution is detected at the adapter ligation step. The library is then subjected to bait-capture and bisulfite conversion. **d**, The size distribution is further shifted upon library amplification to a final library size of 300-600 bp representing DNA fragments of ~150-500 bp.

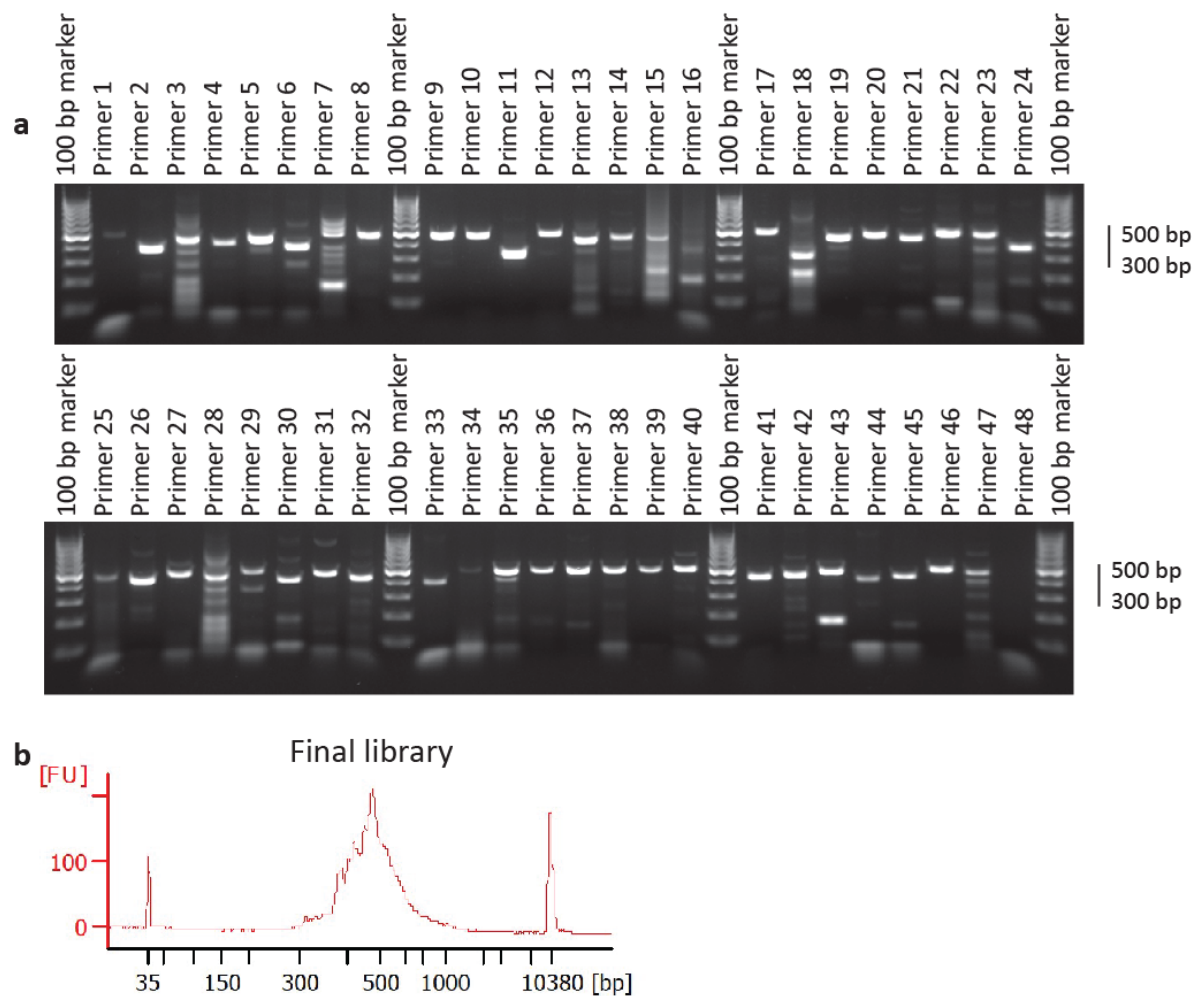


Figure 3: Quality controls during the preparation of the amplicon SMF samples. 1-2 μ g of footprinted DNA is bisulfite converted and used as an input for 96 parallel PCR reactions. **a**, PCR efficiency is checked by loading an aliquot on a 2% agarose gel. With standard bisulfite primer design parameters, 80-90% of the reactions lead to a detectable product and amplicon size ranges between 300-500 bp. An aliquot of each PCR product is pooled and used as an input for sequencing library preparation. **b**, The size distribution of the final library is verified on an Agilent Bioanalyzer, with an expected size of 430-630 bp.

Bioinformatics analysis

SMF data can be interpreted in bulk and at the single molecule level. Bulk level analysis is performed by calculating average methylation using all sequencing reads covering a locus. The generated profiles typically show large footprints at nucleosome occupied sites (~ 150 bp) and shorter footprints at TF bound regions (< 25 bp) (Figure 6). The binding frequency of TFs or nucleosomes in the cell population can be further quantified by analysing the data at single molecule resolution. This allows to determine the proportion of sequenced molecules that show a footprint for a given factor at individual binding sites (Figure 6). These proportions reflect the binding frequencies of either TFs or nucleosomes at a given locus and quantitatively describe the heterogeneity of CRE usage in a cell population. Additionally, the degree of co-occurrence between neighbouring binding events can be quantified.

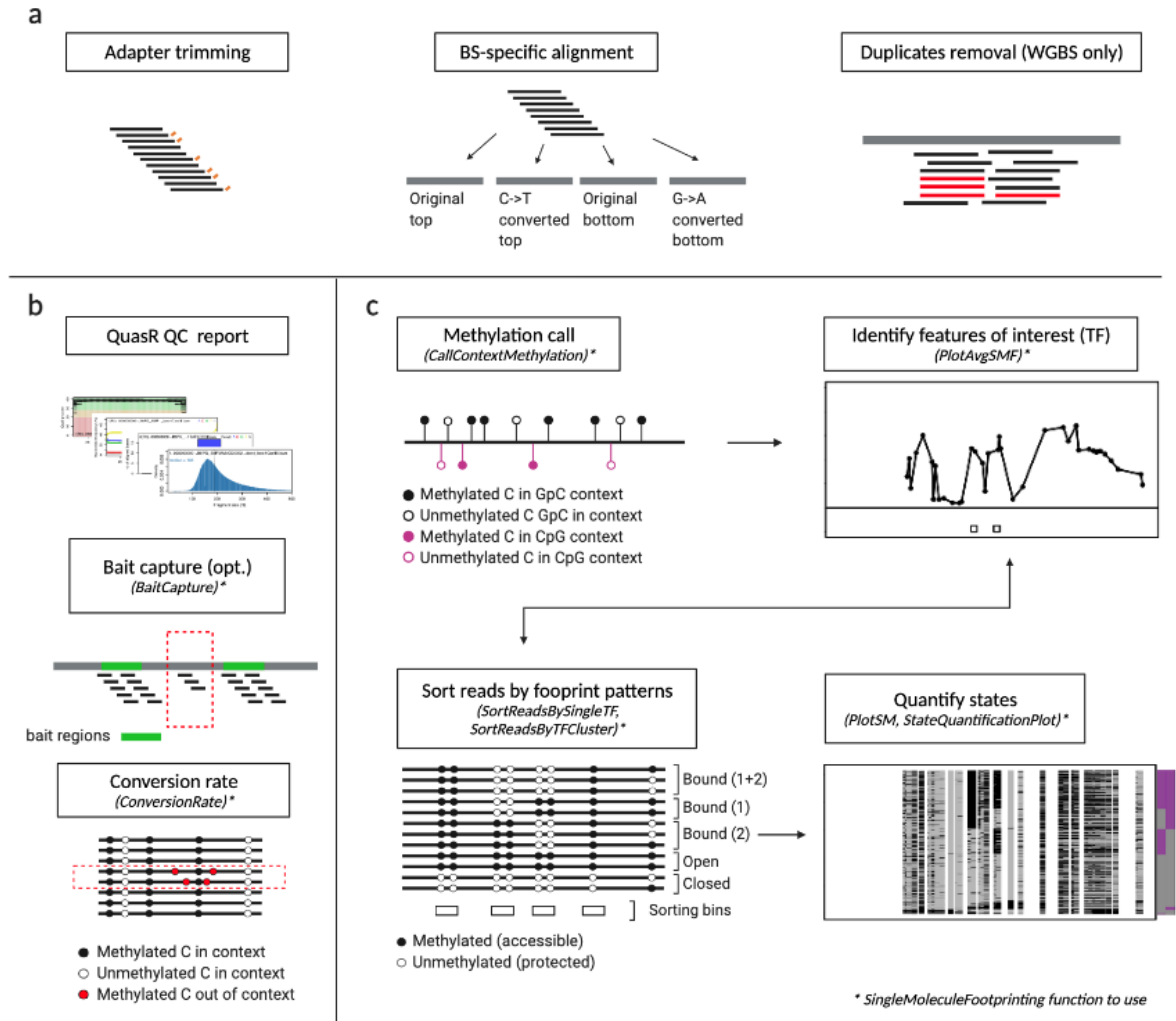


Figure 4. Overview of the computational workflow. **a**, The sequencing reads are pre-processed. Illumina adapters are removed and low-quality bases are trimmed. The reads are aligned against a bisulfite-converted genome and PCR duplicates are removed, only for whole genome bisulfite sequencing experiments (WGBS). **b**, The quality of the library is assessed by performing several generic quality controls including mapping rate, duplication rates, fragment length distribution. In addition, SMF specific controls such as bait capture efficiency and the conversion rate are implemented. **c**, A series of functions have been implemented to ease data interpretation inside the SingleMoleculeFootprinting R package. These include function to call average methylation in the relevant genomic contexts (GpC and CpG); sort the reads according to their footprint patterns; and plot average and single molecule footprints at individual loci.

Data pre-processing

Base calling and barcode demultiplexing of raw Illumina data is performed using manufacturer instructions and software. The resulting fastq are used as an input for adapters trimming and low-quality bases trimming. Reads are aligned using the Bioconductor package QuasR¹⁶ which performs bisulfite alignments using Bowtie 1¹⁷. Using Bowtie 1.x.x rather than later versions (Bowtie 2.x.x) is critical to ensure compatibility with the

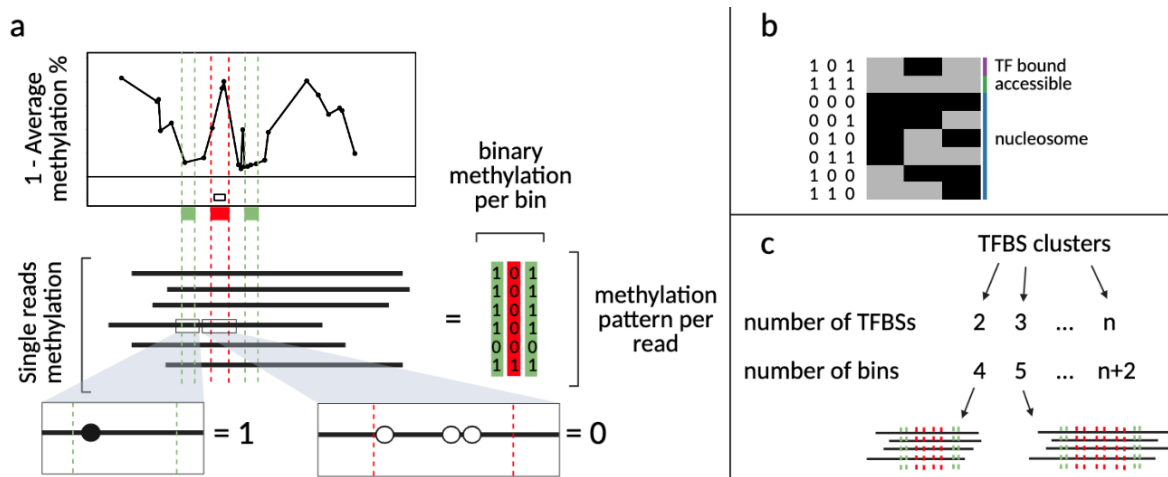


Figure 5. Single molecule sorting. **a**, Single reads can be sorted according to the occupancy pattern over a genomic feature of interest. Here, this is a transcription factor binding site (TFBS) depicted as the white box in the lower part of the average SMF plot. Three collection bins are drawn: one centered on the TFBS (red box), one upstream and one downstream of it (green boxes). For each read, the methylation information is averaged and rounded within the bins (as shown in the callout windows). The result is that each read is now reduced to three binary values. **b**, There are 2^3 possible methylation patterns. One of those is “101” which represents the cases where the TFBS bin is found occupied (unmethylated) and the two surrounding bins are found accessible (methylated). When the methylation pattern of a read corresponds to “101”, it is interpreted as in the “TF bound” state. Alternatively, the sequence “111” would correspond to the “accessible” state. The remaining combinations are interpreted as “nucleosome occupied” states. **c**, Single reads can also be sorted according to the occupancy pattern over multiple genomic features, such as TFBS clusters. In this case, the number of bins that are drawn is $n+2$, where n equals the number of TFBS in the cluster. Notably, the number of possible states, and therefore the complexity of the biological interpretation, increases with the number of TFBSs.

SingleMoleculeFootprinting R package. The expected mapping rate for a typical mouse SMF experiment is ~60%. Technical replicates are pooled and PCR duplicates are removed using Picard Tools v2.15.0¹⁸. Duplicates removal should not be performed in case of amplicon sequencing experiments.

Quality Controls

The *qQCreport* function from *QuasR* provides an assessment of the quality of the SMF libraries. A typical library has a mapping rate of >50%, the read duplication rate is <20% (does not apply to amplicon sequencing experiments) and the fragment size distribution has a median of over 200 bp. In addition, the function *ConversionRateEstimate* from *SingleMoleculeFootprinting* allows the estimate of the bisulfite conversion rate by measuring the conversion of cytosines outside of methylated contexts. Conversion rates should exceed 95%. The efficiency of the optional capture step for large genomes can be calculated using the *BaitCapture* function from *SingleMoleculeFootprinting*. For a typical captured SMF library, more than 70% of the reads are expected to fall within the region targeted by baits.

Quantification of bulk protein occupancy levels

Average methylation is computed at all genomic cytosines and reduced to the relevant contexts using the *SingleMoleculeFootprinting* function *CallContextMethylation*, which employs at its core the *QuasR* function *qMeth*. GpC and CpG contexts can be analysed together when performing dual-enzyme footprinting (e.g., in *Drosophila*). However, they have to be interpreted separately when performing GpC-only footprinting in mammalian cell-lines that have endogenous CpG methylation. In this case, accessibility footprints should only be analysed at DGCHN contexts to exclude ambiguous contexts such as GCGs that are also targeted by endogenous methyltransferases.

Assigning the identity of SMF footprints

As any other footprinting method, SMF is agnostic to the identity of the protein creating the footprints. Thus, SMF data interpretation requires the association of the observed footprint with the protein that creates it. Scanning the genome with known Positional Weight Matrices (PWMs)¹⁹, can be used to annotate the footprints and identify putative TFs binding events. This process is however very noisy, typically leading to multiple overlapping motifs, most of which are not bound. Therefore, we advise to subset the list of putative Transcription Factor Binding Sites (TFBSs) for evidence of *in vivo* binding as returned by orthogonal methods such as ChIP-seq. For instance, UniBind²⁰ offers a reference map of putative Transcription Factor Binding Sites (TFBSs) predicted from ChIP-seq data.

Quantification of protein occupancy at single molecule level

We developed several strategies to sort molecules according to their occupancy states and to calculate the frequency of those states at individual loci (Figure 5). In the case of TFs, we distinguish molecules that are bound by one or multiple TFs, from molecules that are fully accessible and molecules that are occupied by nucleosomes. Given a set of n TFBS coordinates as input, we draw $n+2$ bins: one per TFBS, plus one upstream and one downstream (Figure 5). Methylation values for each read are averaged and rounded within each bin such that each read becomes described by a string of $n+2$ binary digits. There are at this point 2^{n+2} possible methylation patterns that can be biologically interpreted in terms of molecular occupancy. The functions *SortReadsBySingleTF* and *SortReadsByTFCluster* from our *SingleMoleculeFootprinting* package can be used to sort reads based on the footprint left by one or multiple TFs, respectively.

Single locus visualization

The bulk footprinting signal can be displayed for a single locus using the function *PlotAvgSMF*. Accessibility information for single molecules can be visualized using the *PlotSM* function, while the proportions of reads found in each state can be obtained using the *StateQuantificationPlot* function. Finally, the function *PlotSingleSiteSMF* offers a convenience wrapper for the three (Figure 6).

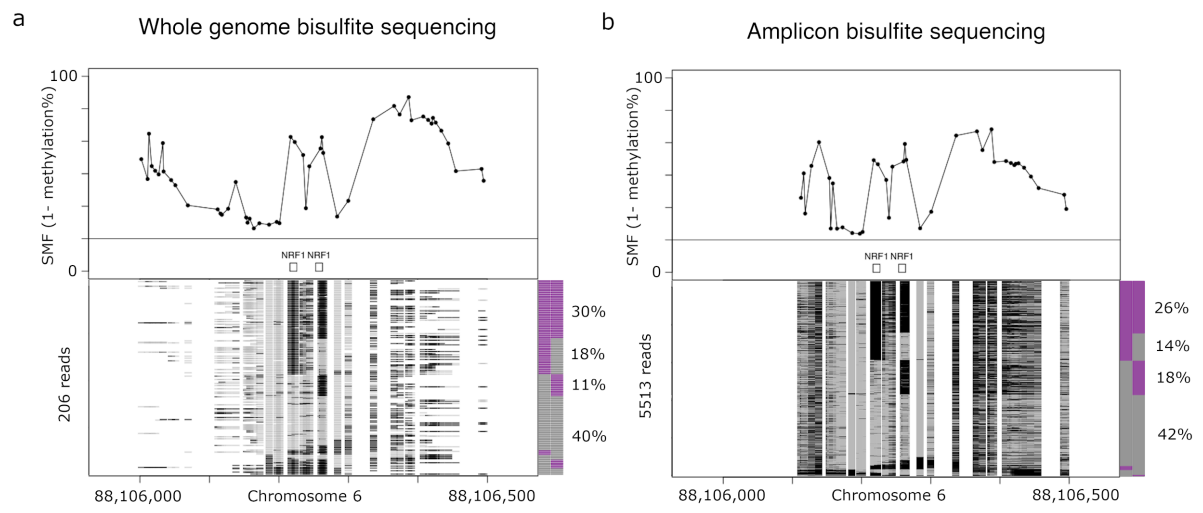


Figure 6. SMF data visualisation. Single molecule analysis of a genomic locus harbouring two NRF1 binding sites using **a**, whole genome bisulfite sequencing (WGBS) or **b**, amplicon bisulfite sequencing data. The upper panels show the average SMF signal (1-methylation). The lower panels show stacks of single DNA molecules sorted according to the occupancy pattern of the two NRF1 binding sites. The frequency of the states is displayed in the barplot next to the single molecule stacks. In this particular case, both NRF1s are found to co-occupy 30% and 26% of the reads in the WGBS and amplicon sequencing experiment, respectively. Binding by individual NRF1 ranges between 11% and 18% of the reads and the region is accessible in about 40% of the molecules. Signal amplification in the amplicon experiment increases coverage to 5513 reads versus the 206 of the genome-wide experiment.

Advantages of the method

Most genomics methods used to measure protein-DNA interactions such as ChIP-seq, CUT&RUN, DNase-seq or ATAC-seq are bulk assays that average binding signals over millions of cells. These are based on the selective sequencing of protein-bound DNA fragments through their fragmentation and enrichment. This enrichment step implies that only the protein-bound DNA molecules are quantified. Potential heterogeneity such as the competition between nucleosomes and TFs would be ignored. Moreover, all these protocols disrupt the chromatin template, thus precluding the measure of how multiple factors interact on DNA. In SMF, molecules are sequenced regardless of their accessibility status. Thus, at any given locus, the competitive occupancy by TFs and nucleosomes can be simultaneously quantified, providing valuable information of the frequency of CRE usage in cellular populations⁸. Moreover, deposition of methylation in SMF preserves the integrity of DNA, allowing to measure the footprints created by multiple proteins over a stretch of 300-500 bp of a DNA molecule. This information can be used to infer dependencies between binding events in the genome and has for instance allowed us to resolve the mechanism of TF cooperativity *in vivo*⁸.

Recently, single cell protocols have been developed for most genomics assays²¹, allowing to resolve the heterogeneity of CRE usage in individual cells. The generated data have sufficient resolution to precisely infer the cell type composition of heterogeneous

populations^{22,23}. However, information per single cell is sparse, and a given CRE rarely has more than a couple of informative reads per cell which is insufficient to dissect the logic of protein binding events at CREs²¹. SMF provides complementary information to single cell approaches as it resolves details of the molecular occupancy patterns at CREs²¹. In turn, SMF comes as a method of choice when dissecting molecular mechanisms regulating transcription.

Limitations of the method

SMF detects protein-DNA contacts through a protection of DNA from the exogenous methylation signal. As any footprinting method, SMF is agnostic to the identity of the protein creating it. Thus, interpretation of the SMF signal requires the integration of other sources of information. For instance, we have demonstrated that combining TF recognition motifs and ChIP-seq can be used to accurately identify the TFs creating footprints detectable by SMF⁸. Similarly, we have shown that footprints created by GTFs and Pol II at core promoters can be identified by their relative position to transcriptional start sites as defined by CAGE data¹⁰. It is therefore recommended to apply SMF to understand the dependencies between binding events for which the identity of the factors and their binding location is documented by orthogonal methods. Moreover, confirming the identity of the factor through downregulation is advisable to unambiguously identify the factors creating the footprints^{8,10}.

SMF uses McviPI and/or SssI that methylate cytosines in GpC and CpG context respectively. This implies that molecular accessibility can only be resolved at regions containing sufficient density of these dinucleotides. The use of methylation in the CpG context is restricted to biological systems where endogenous DNA methylation is absent in this context. This is the case for flies and embryonic stem cells that can proliferate in absence of endogenous DNA methylation. We empirically defined that the footprints created by TFs or Pol II span 15-20 bp^{8,10}. This is compatible with the resolution of SMF performed using either the GpC methyltransferase only (~14 bp) or both enzymes (~7 bp). However, dinucleotide distribution is not even across the genome. For a given genome, only a fraction of the binding sites for every TF (i.e. ~20% for REST⁸) or Pol II pausing sites (~40% in flies¹⁰) will be analysable by SMF. It is therefore important to analyse the dinucleotide compositions of the regions of interest prior to SMF profiling (Figure 7).

DNase-seq or ATAC-seq are based on the selective sequencing of accessible regions of the genome, while SMF sequences all DNA molecules regardless of their accessibility status. Performing SMF is significantly more expensive than other DNA footprinting methods. Costs are in large part attributed to the requirement of high coverage (>40x) for sound statistical analysis of binding frequencies. This problem is enhanced for mammalian genomes that are 20 times bigger than fly genomes. We have developed several strategies for targeted SMF that enable cost-efficient high coverage SMF on hundreds (PCR-based^{8,10}) to tens of thousands of loci (bait capture on mouse CREs⁸). Thereby permitting focus of the sequencing efforts to the regulatory regions that only represents 5-10% of mammalian genomes.

Applications of the method

SMF has been applied to several *Drosophila* and mouse cell lines successfully. Moreover, various protocols for methylation footprinting have been developed and used in yeast^{24,25} and humans^{9,11}. In principle SMF can be adapted to any cell type or tissue for which nuclei can be purified and permeabilized. Efficient footprinting depends on a homogeneous nuclear extract, therefore the protocol may have to be adapted according to the cell type. SMF is performed on purified nuclei under native conditions. An alternative approach consists of performing the methylation footprinting on crosslinked-chromatin. We and others have successfully implemented such a protocol^{8,24,25}. This strategy globally generates comparable results to native SMF⁸. However, the ability to add accessibility information on stable protein-DNA complexes has the potential to enable the coupling of SMF with other approaches such as ChIP or HiC. In turn, such a technology could resolve the genomic and epigenomic context in which TF binding occurs at the molecular level.

Recently, several studies have demonstrated the possibility to couple methylation footprinting with long read sequencing^{12–15}. The advancements made by these studies enable haplotype resolved maps of accessibility over several kilobases. This continuous accessibility information can reveal relations and dependencies between distant CREs. Future improvements of these sequencing methods in terms of throughput and methylation calls accuracy could enable measuring the degree of TF co-occupancy at distant regulatory regions.

Expertise needed to implement the protocol

The protocol described here requires standard molecular biology techniques. Production of single molecule footprinting data requires access to a dedicated sequencing facility. In order to analyse SMF data, the user should have access to a high-performance computing system with a Linux distribution installed in order to perform some of the data pre-processing steps. The user should be comfortable with R scripting and minimal command line usage.

MATERIALS

Reagents

Common reagents

- Agencourt AMPure XP magnetic beads (Beckman Coulter, cat. no. A63880)
- Cell suspension. Mouse ES cells; 159²⁶ and 159 DNMT TKO²⁷
- Chloroform (Sigma, cat. no. 366919)
- CAUTION: harmful and irritant, avoid inhalation, wear personal protective equipment (PPE).
- CpG Methyltransferase (SssI) (NEB, cat. no M0226L)
- DMEM high glucose (Gibco, cat. No. 41965039)
- Ethanol (Sigma, cat. No. 1.00983)
- CAUTION: flammable
- FBS Embryomax (Millipore, cat. No. ES-009-B)
- Glycogen from *Mytilus edulis* (Blue mussel) (Sigma-Aldrich, cat. no. G1767) (stock solution 20mg/ml)
- GpC Methyltransferase (M.CviPI) (NEB, cat. no M0227L)
- IGEPAL CA-630 (Sigma, cat. No. I8896)
- CAUTION: eye irritant
- L-glutamine (Gibco, cat. no A2916801)
- LIF (in house 10ug/ml)
- Magnesium chloride (MgCl₂) (Sigma, cat. No M8266)
- MEM Non-Essential Amino Acids Solution (100X) (Gibco, cat. No. 11140050)
- 2-Mercaptoethanol (Merck, cat. no M6250)
- CAUTION: Toxic and irritant, avoid inhalation wear PPE
- Sodium pyruvate (Gibco, cat. No. 11360070)
- Nuclease-Free Water (not DEPC-Treated) (Ambion, cat.no. AM9937)
- Phenol equilibrated, stabilized : Chloroform : Isoamyl Alcohol 25 : 24 : 1 (PCI) (PanReacAppliChem, cat. no. A0889)
- CAUTION: phenol is corrosive and toxic, chloroform is harmful and an irritant. Avoid inhalation and wear PPE.
- Phosphate buffered saline solution (PBS) (in house)
- 2-propanol (Sigma, cat. No. I9516)
- CAUTION: flammable
- Proteinase K (Sigma-Aldrich, cat. no 124568) (working solution 20mg/ml)
- Qubit dsDNA HS Assay Kit (Life Technologies, cat. no. Q32851)
- RNase A, DNase- and protease-free (10 mg/ml) (Sigma, cat. no. R6513)
- S-adenosyl-methionine (SAM) (32 mM) (NEB, cat. no B9003S)
- Sodium chloride (NaCl) (Sigma-Aldrich, cat. no. S7653)

- Sodium dodecyl sulfate solution (10%) (SDS) (Sigma-Aldrich, cat. no. 71736)
- Sucrose Ultrapure MB grade (Affymetrix, cat. No. 21938)
- Titriplex III (ethylenedinitrilotetraacetic acid disodium salt dihydrate) (EDTA) (Sigma, cat. no. 1.08421)
- Trypan blue solution (0.4% (wt/vol)) (Gibco, cat. no. 15250061)
- Trypsin-EDTA (0.25%) (Gibco, Cat. No. 25200056)
- Trizma base (Sigma-Aldrich, cat. No. T1503)

Capture library

- EZ DNA-methylation gold kit (Zymo research, cat. No. D5005)
- Sodium hydroxide solution (10 M) (Sigma-Aldrich, cat. No. 72068)
- CAUTION: Corrosive, wear PPE.
- SureSelectXT Methyl-Seq Reagent Kit (Agilent, cat. no. G9651A)
- SureSelectXT Mouse Methyl-Seq Capture Library (Agilent, cat. no. 931052)

Amplicon library

- Agarose (Sigma, cat. No. A9539)
- Epitect bisulfite conversion kit (Qiagen cat. No. 59104)
- Ethidium bromide solution 1 % (Roth, cat. No. 2218.1)
- GeneRuler 1 kb DNA Ladder, ready-to-use (Thermo, cat. No. SM0313)
- GeneRuler 100 bp DNA Ladder, ready-to-use (Thermo, cat. No. SM0244)
- KAPA HiFi HotStart Uracil+ ReadyMix (2X) (Roche cat. No. KK2802 07959079001)
- NEBNext Ultra II DNA Library Prep Kit for Illumina (NEB, cat. No. E7645L)
- NEBNext Multiplex Oligos for Illumina (Index Primers 1-12) (NEB, cat. No. E7335L)
- Primers, bisulfite specific, resulting in amplicons ranging from 300-500 bp (Sigma)

Equipment

Common

- Bioanalyzer with chip for DNA 1000 and DNA HS (Agilent)
- Centrifuge, refrigerated, with fixed-angle rotor (Eppendorf, model no. 5427R)
- Centrifuge with fixed-angle rotor (Eppendorf, model no. 5425)
- Centrifuge with swinging bucket (Eppendorf, model no. 5810R)
- Heater block with wells for 1.5-ml tubes (set to 37 and 56 °C)
- Magnetic rack for PCR tubes (in house)
- Magnetic rack for 1.5-ml tubes; Dynamag (Thermo)
- Microcentrifuge (e.g., Roth)
- 1.5-ml Microcentrifuge tubes (Eppendorf, cat. no. 22-282)
- 1.5-ml Microcentrifuge Safe-Lock tubes (Eppendorf)

- 1.5-ml Microcentrifuge DNA LoBind tubes (Eppendorf)
- 0.2-ml PCR tubes (Eppendorf)
- Thermal cycler (Biorad, C1000 touch, 1851148/1851196)
- Vortex mixer (e.g., Vortex Genie; VWR)
- Water baths (set to 37°C)

Capture library

- MicroTUBE holder (Covaris, cat. no. 500114)
- S-series focused ultrasonicator (Covaris)
- Snap-Cap microTUBEs (Covaris, cat. no. 520045)
- Vacuum concentrator (e.g. Eppendorf)

Amplicon library

- Agarose gel chamber and power supply (Bio-rad, Cat no. 1640301)
- PCR adhesive film (Eppendorf, Cat no. 0030127781)
- Twin.tec PCR plates (Eppendorf, Cat no. 0030133366)

Reagent setup

Lysis Buffer Recipe: 10 mM Tris (pH 7.4), 10 mM NaCl, 3 mM MgCl₂, 0.1 mM EDTA, 0.5% IGEPAL CA-630. Buffer is stable for up to 1 month at 4°C.

Wash Buffer Recipe: 10 mM Tris (pH 7.4), 10 mM NaCl, 3 mM MgCl₂, and 0.1 mM EDTA. Buffer is stable for up to 1 month at 4°C.

GpC Methyltransferase Mix: 1X M.GpC buffer, 300 mM Sucrose and 64 µM SAM. Add SAM shortly before using the mix.

Stop solution: 20 mM Tris-HCl, pH 7.9, 600 mM NaCl, 1% SDS and 10 mM EDTA. Needs to be heated up to 37°C prior to usage to eliminate precipitates.

PROCEDURE

Nuclei extraction

NOTE: For downstream application of genome wide footprinting generally 3 µg of DNA is needed, therefore 3 reactions should be performed. Per reaction 0.25 10^6 mESCs cells are needed.

1. Actively growing cells are trypsinized and washed once with cold PBS.

CRITICAL STEP: The following steps are all done at 4°C. The lysis and wash buffer are also kept at 4°C. Prewarm the stop solution at 37°C.

CRITICAL STEP: Here, it is important to note that a single cell suspension is essential in this step, since a mistake can only be read out upon data analysis. Therefore, confirm single cell distribution under a microscope or with a cell counter.

2. Resuspend 0.25 10^6 cells in 1 ml of ice-cold lysis buffer, incubate on ice for 5-10 min, invert the tubes occasionally.
3. Centrifuge for 5 min at 1000x g at 4°C. The supernatant is discarded.
4. Nuclei are washed with 250 µl ice-cold wash buffer.
5. Centrifuge for 5 min at 1000x g at 4°C. The supernatant is discarded.
6. The nuclei are then resuspended in 94.5 µl 1X M.GpC buffer and kept on ice.

GpC methyltransferase treatment

7. To each sample containing 94.5 µl of nuclei add 150 µl of GpC methyltransferase mix.

CRITICAL STEP: It is important to be fast from this step on, in order to reduce time-induced treatment bias.

8. Add 50 µl of M.cviPI and mix by pipetting with P200. Do not vortex.
9. Incubate at 37°C for 7.5 min.

CRITICAL STEP: Keep samples at room temperature (RT) for the following additions.

10. Add 25 µl of M.cviPI, then 4 µl SAM and mix by pipetting with P200. Do not vortex.
11. Incubate at 37°C for 7.5 min.

CpG methyltransferase treatment

NOTE: CpG methyltransferase treatment (optional, only for cells without endogenous methylation, such as DNMT TKO).

CRITICAL STEP: Keep samples at RT for the following additions.

12. Add in this order; 3.5 µl MgCl₂ (1M), 15 µl SssI and 4 µl SAM. Mix by pipetting with P200. Do not vortex.

NOTE: Leftover SAM will not be reused.

13. Incubate at 37°C for 7.5 min.

Finalize treatment

14. Add 300 µl of pre-warmed stop solution, 6 µl proteinase K and mix briefly by vortex.

15. Incubate overnight at 55°C.

PAUSE POINT: samples can be frozen at this point

DNA extraction

16. Extract DNA by adding 600 µl PCI to each sample.

17. Shake hard 15 times and centrifuge 5 min at maximum speed.

18. Transfer the aqueous phase to a new 1.5-ml safelock tube.

19. Add 600 µl Chloroform, shake hard 15 times and centrifuge 5 min at maximum speed.

20. Transfer the aqueous phase to a new 1.5-ml safelock tube.

21. Precipitate the DNA by adding 600 µl Isopropanol and 1 µl glycogen.

22. Incubate at RT shaking continuously for 10 min.

23. Centrifuge 20 min at maximum speed.

24. Remove the supernatant.

25. Wash pellet with ice-cold 70% ethanol.

26. Centrifuge 15 min at 4°C at maximum speed.

27. Remove the supernatant thoroughly without disturbing the pellet.

28. Air dry pellet by laying the tubes on the bench.

29. Resuspend pellet in 20 µl H₂O for amplicon application and in 30 µl H₂O for target enrichment application.

30. Add 1 µl RNase A and incubate at 37°C for 30 min.

31. Let pellet dissolve fully for 2hrs at 37°C or overnight at 4°C.

PAUSE POINT: samples can be frozen at this point.

32. Quantify DNA concentration by Qubit 1X DNA HS measurement and DNA quality by Nanodrop.

NOTE: Expect ~1 µg DNA per reaction

Application A: Whole genome bisulfite sequencing library with targeted enrichment

NOTE: Libraries are prepared based on the SureSelect XT Mouse Methyl-Seq Kit Enrichment System for Illumina Multiplexed Sequencing Library protocol (Agilent Technologies, Santa Clara CA, Version E0, April 2018)

NOTE: Samples are fragmented with a Covaris model S2. Do consult the S-series setup and instruction manual for start-up procedures (Covaris, Chapter 4.1, Rev F October 2020)

Library preparation

NOTE: Prepare the Covaris device ahead of time as this will take time to cool down.

NOTE: Do not keep the DNA in microtube for longer times, add it shortly before fragmentation.

33. Prepare a dilution of 3 µg DNA in 130 µl in a microtube.
34. Fragment DNA using a Covaris device. Aim to obtain 300 bp fragments via sonication. For Covaris S2 this will be duty factor 10%, intensity 4 and 200 cycles/burst for 100 sec. Check fragmentation quality with bioanalyzer by running 1 µl of 1 in 5 diluted sample. A successful example is shown in figure 2a.

? TROUBLESHOOTING

Reagents for End Repair master mix	Volume for 1 rxn
10x End Repair Buffer (clear cap)	10 µl
dNTP Mix (green cap)	1.6 µl
Klenow DNA Polymerase (yellow cap)	2 µl
T4 Polynucleotide Kinase (orange cap)	2.2 µl
T4 DNA Polymerase (purple cap)	1 µl
<i>Total</i>	<i>52 µl</i>

35. Prepare the end repair master mix, keep on ice.
36. Concentrate fragmented DNA down to final volume 83.2 µl with a vacuum concentrator for 20 min at 30°C.
37. Add the end repair master mix to the concentrated DNA. Mix briefly by vortex and spin for a few sec on table top spinner. Run the sample with the following program on a thermal cycler.

Program setup	Temperature	Time
Step 1	20°C	30 min
Step 2	4°C	Hold

38. Add 180 µl of AMPure XP beads to 100 µl of end repaired sample (1.8x ratio) and mix by pipetting for about 10 times.

CRITICAL STEP Prior to usage put AMPure XP beads at RT. To ensure a correct ratio is maintained, mix the beads well on a vortex shortly before adding them to the sample.

39. Incubate sample for 5 min at RT.

40. Place the sample on a magnetic stand until the solution is clear. This will take about 5 min.

41. While keeping the tubes on the magnetic stand remove the supernatant.

42. Wash the beads with 200 µl freshly prepared 80% EtOH, while still keeping the sample on the magnetic stand.

CRITICAL STEP: Freshly prepared EtOH should not be older than 48 hrs as this will reduce the concentration of the washing solution and in effect the output of the sample.

43. Repeat washing step 42 once more.

44. After the second wash, remove the supernatant complete and lay the tubes on their side to air dry the beads. This will take approximately 1-3 min.

CRITICAL STEP: Air drying the beads is very critical. Avoid over drying the beads, as this would mean a substantial loss of material when rehydrating the beads.

45. Once the beads are no longer glossy add 44 µl water and resuspend the dried pellet by tapping the tube.

46. Incubate for about 2 min.

47. Place the sample on a magnetic stand until the solution is clear. This will take about 2 min.

48. Transfer 42 µl of eluate to a new tube.

49. Prepare an aliquot (1 µl, 4 x diluted) to check the quality with a bioanalyzer DNA HS chip later

NOTE: This will be run later with ligated sample, step 60. A successful example is shown in figure 2b.

50. Prepare the A-tailing master mix, keep on ice.

Reagents for A-tailing master mix	Volume for 1 rxn
10× Klenow Polymerase Buffer (blue cap)	5 μ l
dATP (green cap)	1 μ l
Exo(–) Klenow (red cap)	3 μ l
<i>Total</i>	<i>9 μl</i>

51. Add 9 μ l prepared A-tailing mix to 41 μ l purified end-prepped DNA, mix briefly by vortex and spin down for a few seconds on a table top spinner. Run the sample with the following program on a thermal cycler.

Program setup	Temperature	Time
Step 1	37°C	20 min
Step 2	4°C	Hold

52. Clean up the A-tailed sample by adding 90 μ l of AMPure XP beads to 50 μ l of the sample (1.8x ratio).

53. Follow steps 38-48 as described earlier in protocol to carry out the clean-up.

54. Elute sample by adding 35 μ l water and transfer 33.5 μ l of eluate to a new tube.

55. Prepare the Ligation master mix, keep on ice.

Reagents for Ligation master mix	Volume for 1 rxn
SureSelect Methyl-Seq Methylated Adapter (green cap)	5 μ l
5× T4 DNA Ligase Buffer (green cap)	10 μ l
T4 DNA Ligase (red cap)	1.5 μ l
<i>Total</i>	<i>16.5 μl</i>

56. Add 16.5 μ l ligation master mix to 33.5 μ l purified A-tailed DNA, mix briefly by vortex and spin down. Run the sample with the following program on a thermal cycler.

Program setup	Temperature	Time
Step 1	20°C	15 min
Step 2	4°C	Hold

57. Clean up the ligated sample by adding 32.5 µl of AMPure XP beads to 50 µl of the sample (0.65x ratio).
58. Follow steps 38-48 as described earlier in protocol to carry out the clean-up.
59. Elute the sample by adding 24 µl water and transfer 22 µl of eluate to a new tube.
60. Check quality with a bioanalyzer DNA HS chip (1 µl, 4 x diluted), also run the aliquot from step 49, a successful example is shown in figure 2c. In addition, quantify the ligated sample with Qubit 1x DNA HS.

NOTE: if less than 350 ng of adapter ligated DNA is recovered, repeat the library preparation to obtain more material. Expect 500-800 ng adapter ligated material, when starting with 3 µg footprinted DNA.

PAUSE POINT: samples can be frozen at this point.

Hybridization and capture of the library

NOTE: At this point it is good to keep in mind that the hybridization step takes 16 hrs. Remember when planning this part of the experiment that there is no pause point in the remainder of the protocol.

61. With a vacuum concentrator reduce the adapter ligated DNA down to 3.4 µl. Prepare a test tube with 3.4 µl water as comparison. When reduced to a volume less than 3.4 µl add up to the correct amount with water, but do not mix.

CRITICAL STEP: It is important not to mix the sample with a pipette, since this can result in loss of material.

62. Prepare the blocking mix and add 5.6 µl to the 3.4 µl concentrated adapter ligated library. Mix the reaction mixture gently with a pipette and run sample with the following program on a thermal cycler.

Reagents for Blocking mix	Volume for 1 rxn
Indexing Block 1 (green cap)	2.5 μ l
Block 2 (blue cap)	2.5 μ l
Methyl-Seq Block 3 (brown cap)	0.6 μ l
<i>Total</i>	<i>5.6 μl</i>

Program setup	Temperature	Time
Step 1	20°C	5 min
Step 2	65°C	2 min
Step 3	65°C	Hold

63. While the samples are at 65°C in the thermal cycler prepare the Methyl-Seq Capture Library Hybridization Mix. Keep the RNase blocking solution on ice and the other mixtures at RT.

Reagents for RNase blocking solution	Volume for 1 rxn
RNase Block (purple cap)	0.5 μ l
Nuclease-free water	1.5 μ l
<i>Total</i>	<i>2 μl</i>

Reagents for Hybridization buffer	Volume for 1 rxn
Hyb 1 (orange cap)	6.63 μ l
Hyb 2 (red cap)	0.27 μ l
Hyb 3 (yellow cap)	2.65 μ l
Hyb 4 (black cap)	3.45 μ l
<i>Total</i>	<i>13 μl</i>

Reagents for Hybridization Mix	Volume for 1 rxn
Hybridization buffer	13 μ l
RNase blocking solution	2 μ l
Mouse Methyl-Seq Capture Library	5 μ l
<i>Total</i>	<i>20 μl</i>

64. Keep the PCR tube containing the DNA library with the blocking mix at 65°C in the thermal cycler while adding 20 μ l of the Capture Library Hybridization Mix. Gently mix the reactions by pipette.

65. Incubate the hybridization mixture for 16 hrs at 65°C with a heated lid set to 105°C.

66. Resuspend the Dynabeads MyOne Streptavidin T1 on a vortex mixer.

NOTE: For one hybridization sample 50 μ l of the magnetic bead suspension is needed. In case of multiple samples use 1.5-ml SafeLock tube to prepare beads (step 67-70) for the capture.

67. Wash 50 μ l streptavidin beads with 200 μ l SureSelect Binding Buffer, mix the beads by pipetting up and down 10 times.

68. Place the sample on a magnetic stand until the solution is clear, then remove and discard the supernatant.

69. Repeat steps 67 and 68 two more times.

70. Resuspend the washed beads in 200 μ l of SureSelect Binding Buffer.

71. Keep the PCR tube with the hybridization reaction at 65°C while transferring the entire volume of the hybridization mixture to the PCR tube containing the 200 μ l of washed streptavidin beads. Slowly pipet up and down until the beads are fully resuspended.

72. Cap the tube and seal with parafilm. Then incubate the capture reaction by putting the PCR tube on a vortex mixing continuously at full speed for 30 min at RT. Make sure the sample is properly mixing in the tube.

73. During the 30 min incubation for capture, pre-warm Wash Buffer 2 at 65°C by placing 200 μ l aliquots of Wash Buffer 2 in PCR tubes. Aliquot 3 wells of buffer for each DNA capture sample.

74. Place the aliquots with Wash Buffer 2 in the thermal cycler, with the heated lid ON, held at 65°C.

75. After the 30 min incubation period, briefly spin the capture reaction tube in a centrifuge.

76. Place the sample on a magnetic stand until the solution is clear, then remove and discard the supernatant.

77. Resuspend the beads in 200 μ l of SureSelect Wash Buffer 1 and mix by pipetting until the beads are fully resuspended.

78. Incubate the sample for 15 min at RT. Afterwards briefly spin in a centrifuge.

NOTE: During the 15 min incubation, prepare fresh 0.1 M NaOH to elute the captured library from the beads. (Prepare 20 µl per sample (plus excess) by diluting 10 M NaOH stock solution with nuclease-free water.)

79. Place the sample on a magnetic stand until the solution is clear, then remove and discard the supernatant.
80. Wash the beads with 200 µl of 65°C prewarmed Wash Buffer 2. Pipette up and down until beads are fully resuspended.
81. Cap the wells, then incubate the sample for 10 min at 65°C on the thermal cycler.
82. Place the sample on a magnetic stand until the solution is clear, then remove and discard the supernatant.
83. Repeat washing steps 80-82 twice more. Make sure all of the wash buffer has been removed during the final wash.
84. To elute the captured DNA, add 20 µl of the freshly prepared 0.1 M NaOH solution to the bead-bound sample and mix on a vortex mixer for 5 sec to resuspend the beads. Then incubate the sample for 20 min at RT.

NOTE: During the 20 min incubation, prepare the EZ DNA Methylation-Gold Kit CT Conversion Reagent.

85. Collect the beads from the elution mixture by placing the sample on a magnetic stand for about 2 min. Transfer the eluate, containing the captured DNA, to a new PCR tube.

Bisulfite conversion step

NOTE: Captured libraries are converted with the ZYMO EZ DNA Methylation-Gold Kit according to the manufacturer's protocol.

86. Prepare the CT conversion reagent mix by reconstituting one vial of solid CT Conversion Reagent with 900 µl of nuclease-free water, 300 µl of M-Dilution Buffer, and 50 µl of M-Dissolving Buffer.

NOTE: Prepare the appropriate number of vials for the number of samples in the run. One vial is sufficient for 10 samples.

87. Mix by continuous vortexing for 10 min at RT.
88. Add 130 µl of the prepared CT Conversion Reagent to the 20 µl of captured library sample. Mix by brief vortexing, then briefly spin in a centrifuge.
89. Divide the bisulfite conversion reaction over two PCR tubes. Place the tubes in a thermal cycler and run the following program.

Program setup	Temperature	Time
Step 1	64°C	2.5 hr
Step 2	4°C	Hold

NOTE: When hybridization and capture are omitted (steps 61-85) an additional step at 98°C for 10 min is required prior to step 1 in the program of step 89.

90. Combine the two 75 µl bisulfite conversion reactions to get a total volume of 150 µl for each DNA library.

NOTE: Before starting the desulphonation procedure, make sure that the ethanol has been added to the M-Wash buffer provided with the EZ DNA Methylation-Gold Kit, according to the kit instructions.

91. First add 600 µl of M-Binding Buffer to a Zymo-Spin IC column and place the column in a collection tube. Then load the 150 µl bisulfite converted DNA sample onto the column.

92. Cap the column and mix well by inverting the column five times. Spin down for 1 min at 17,000x g. Discard the flow through, then place the column back in the same collection tube.

93. Wash the column by adding 100 µl of prepared M-Wash Buffer. Spin down for 1 min at 17,000x g. Discard the flow-through, then place the column back in the same collection tube.

94. Add 200 µl of M-Desulphonation Buffer to the column. Incubate at RT for 20 min.

95. Spin down for 1 min at 17,000x g. Discard the flow-through, then place the column back in the same collection tube.

96. Add 200 µl of prepared M-Wash Buffer to the column. Spin down for 1 min at 17,000x g. Discard the flow-through, then place the column back in the same collection tube.

97. Repeat washing step 96 once more.

98. Spin down once more for 1 min at 17,000x g.

99. Place the column in a fresh 1.5-ml tube. Allow the column to sit at RT for 2 min.

100. Add 10 µl of M-Elution Buffer to the column and incubate at RT for 3 min.

101. Spin down for 1 min at 17,000x g.

102. Keep the flow-through in the collection tube and add an additional 10 µl of M-Elution Buffer to the column. Incubate at RT for 3 min.

103. Spin down for 1 min at 17,000x g and continue with the combined eluate for further processing.

104. To amplify the bisulfite converted library, prepare the PCR master mix, keep on ice.

Reagents for PCR master mix	Volume for 1 rxn
Nuclease free water	30 µl
SureSelect Methyl-Seq PCR Master Mix	50 µl
Methyl-Seq PCR1 Primer F	1 µl
Methyl-Seq PCR1 Primer R	1 µl
<i>Total</i>	<i>82 µl</i>

105. Add 82 μ l PCR master mix to 18 μ l bisulfite converted library, mix briefly by vortex and spin down. Run the sample with the following program on a thermal cycler.

Program setup	# cycles	Temperature	Time
Step 1	1	95°C	2 min
Step 2	8	95°C	30 sec
		60°C	30 sec
		72°C	30 sec
Step 3	1	72°C	7 min
Step 4	1	4°C	Hold

106. Clean up the amplified bisulfite converted library by adding 180 μ l of AMPure XP beads to 100 μ l of the sample (1.8x ratio).

107. Follow steps 38-48 as described earlier in protocol to carry out the clean-up.

108. Elute sample by adding 22 μ l water and transfer 19.5 μ l of eluate to a new tube.

Library indexing

109. Prepare the indexing PCR master mix, keep on ice.

NOTE: Assign the indexing barcodes in such a way that optimal diversity is guaranteed. Consult the indexing list in the manufacturer's protocol.

Reagents for indexing PCR master mix	Volume for 1 rxn
SureSelect Methyl-Seq PCR Master Mix	25 μ l
SureSelect Methyl-Seq Indexing Primer Common	0.5 μ l
<i>Total</i>	<i>25.5 μl</i>

110. Add 25.5 μ l indexing PCR master mix to 19.5 μ l amplified bisulfite converted library.

111. Finally, also add 5 μ l selected indexing primer, mix briefly by vortex and spin down. Run the sample with the following program on a thermal cycler.

Program setup	# cycles	Temperature	Time
Step 1	1	95°C	2 min
Step 2	6	95°C	30 sec
		60°C	30 sec
		72°C	30 sec
Step 3	1	72°C	7 min
Step 4	1	4°C	Hold

112. Clean up the final library by adding 90 µl of AMPure XP beads to 50 µl of the sample (1.8x ratio).
113. Follow steps 38-48 as described earlier in protocol to carry out the clean-up.
114. Elute the sample by adding 26 µl water and transfer 24 µl of eluate to a new tube.
115. Check the quality with a bioanalyzer DNA HS chip and the quantity with Qubit 1x DNA HS. A successful example is shown in figure 2d.
116. Run the sample on an Illumina sequencing platform, a Miseq 150 bp paired-end run gives an indication to the quality of the library. A good library can then be run on a Nextseq High 150 bp paired-end.
117. Go to step 93. of Application B for the computational analysis of the sequencing data.

? TROUBLESHOOTING

Application B: Amplicon bisulfite sequencing library

Bisulfite conversion of footprinted DNA

NOTE: Footprinted DNA is converted with the Qiagen Epitect bisulfite kit based on the manufacturer's protocol with some modifications.

NOTE: Before starting the desulphonation procedure, make sure that the ethanol has been added to the buffer BD and BW provided with the Epitect bisulfite kit, according to the kit instructions.

33. Dissolve the required number of aliquots of bisulfite mix by adding 800 µl RNase-free water to the aliquot. Vortex until the bisulfite mix is completely dissolved. This can take up to 5 min.

NOTE: If necessary, heat the bisulfite mix–RNase-free water solution to 60°C and vortex again.

CRITICAL: Do not place dissolved bisulfite mix on ice.

34. Prepare bisulfite reaction by adding 85 μ l bisulfite mix and 35 μ l DNA protect buffer to 20 μ l footprinted DNA in a PCR tube.
35. Run the sample with the following program on a thermal cycler.

Steps	Time	Temperature
Denaturation	5 min	95°C
Incubation	25 min	60°C
Denaturation	5 min	95°C
Incubation	85 min (1 h 25 min)	60°C
Denaturation	5 min	95°C
Incubation	175 min (2 h 55 min)	60°C
Hold	Indefinite	20°C

36. Next, briefly centrifuge the PCR tubes containing the bisulfite reactions and transfer to new 1.5-ml tubes.

NOTE: Transfer of precipitates in the solution will not affect the performance or yield of the reaction.

37. Add 560 μ l buffer BL to the sample. Mix the solutions by vortexing and then centrifuge briefly.
38. Place the necessary number of EpiTect spin columns and collection tubes in a suitable rack. Transfer the mixture from step 37 into the corresponding EpiTect spin column.
39. Centrifuge the spin columns at maximum speed for 1 min. Discard the flow-through, and place the spin columns back into the collection tubes.
40. Add 500 μ l buffer BW to the spin column and centrifuge at maximum speed for 1 min. Discard the flow-through and place the spin columns back into the collection tubes.
41. Add 500 μ l buffer BD to the spin column, and incubate for 15 min at RT.

CRITICAL STEP: If there are precipitates in buffer BD, avoid transferring them to the spin columns. The bottle containing buffer BD should be closed immediately after use to avoid acidification from carbon dioxide in the air. It is important to close the lids of the spin columns before incubation.

42. Centrifuge the spin columns at maximum speed for 1 min. Discard the flow-through, and place the spin columns back into the collection tubes.
43. Add 500 μ l buffer BW to the spin column and centrifuge at maximum speed for 1 min. Discard the flow-through and place the spin columns back into the collection tubes.

44. Repeat step 43 once more.
45. Place the spin columns into new 2 ml collection tubes, and centrifuge the spin columns at maximum speed for 1 min to remove any residual liquid.
46. Place the spin columns with open lids into new 1.5-ml tubes and incubate the spin columns for 5 min at 56°C in a heating block.
47. Place the spin columns into new 1.5-ml tubes. Dispense 200 µl buffer EB onto the centre of the membrane.
48. Incubate for 5 min at RT. Elute the purified DNA by centrifugation for 1 min at approximately 15,000 x g.

PAUSE POINT: samples can be frozen at this point.

Plate-based PCR for amplicon generation

Box- Primers for amplicon bisulfite sequencing library

Products ranging from 300 to 500 bp in size with the majority of amplicons being over 450 bp is a desired amplicon size distribution.

For compatibility, primers are ordered in 96-mixed well format, where forward and reverse primers for a particular target are present in a single well at 100 µM. Prepare a working plate by diluting primers to 2 µM with RNase-free water in a 96-well format, in a total volume of 200 µl. Working plates can be stored at 4°C, while stock plates are kept at -20°C.

CRITICAL STEP: Pipetting in this section must be performed in a PCR hood to minimize the risk of cross contamination.

49. For the sample, a PCR plate is prepared by aliquoting 5 µl of each primer from the working plate to a new 96-well plate.
50. Prepare a PCR master mix and add 11 µl per well to the PCR plate with the aliquoted primer mix.

Reagents for PCR master mix	Volume for 110x mastermix
PCR-grade water	130 µl
2X KAPA HiFi HotStart Uracil+ ReadyMix	880 µl
Bisulfite converted DNA	200 µl

51. Run the sample plate with the following program on a thermal cycler:

Program setup	# cycles	Temperature	Time
Step 1	1	95°C	3 min
Step 2	35	98°C	20 sec
		56°C	30 sec
		72°C	1 min
Step 3	1	72°C	5 min
Step 4	1	4°C	Hold

PAUSE POINT: samples should be frozen at -20°C when not directly proceeding with DNA purification.

52. To verify product quality and quantity 5 µl of each sample is run of a 2% TBE gel. Figure 3a shows an example where the desired amplicons range from 300-500 bp.

NOTE: For bisulfite specific primers, generally an 80% success rate is expected after conversion. When first using a new set of amplicon primers, it is advisable to run aliquots of the full plate on a gel. For subsequent experiments, it would be sufficient to check only a few amplicons.

53. Upon confirmation by gel, pool 10 µl of each reaction together.

NOTE: Use a multichannel and a PCR strip tube for convenience.

54. Take 800 µl of the pooled sample, add 640 µl of AMPure XP beads (0.8X ratio) and mix by pipetting about 10 times.

CRITICAL STEP: Prior to usage put AMPure XP beads at RT. To ensure a correct ratio is maintained mix the beads well on a vortexer shortly before adding them to the sample.

55. Incubate the sample for 5 min at RT.

56. Place the sample on a magnetic stand until the solution is clear. This will take about 5 min.

57. While keeping the tubes on the magnetic stand, remove and discard the supernatant.

58. Wash the beads with 200 µl freshly prepared 80% EtOH, while still keeping the sample on a magnetic stand.

CRITICAL STEP: Freshly prepared EtOH should not be older than 48 hrs as this will reduce the concentration of the washing solution and in effect the output of the sample.

59. Repeat washing step 58 once more.

60. After the second wash, remove the supernatant completely and lay the tubes on their side to air dry the beads. This will take approximately 1-3 min.

CRITICAL STEP: Air drying the beads is very critical. Avoid over drying the beads, as this would mean a substantial loss of material when rehydrating the beads.

61. Once the beads are no longer glossy add 52 µl water and resuspend the dried pellet by tapping the tube.

62. Incubate for 2 min.

63. Place the sample on a magnetic stand until the solution is clear. This will take about 2 min.

64. Transfer 50 µl of eluate to a new PCR tube.

PAUSE POINT: sample can be frozen at -20°C at this point.

Library preparation

NOTE: Libraries are prepared based on NEBNext DNA Ultra II library preparation protocol.

NOTE: Up to 12 amplicon bisulfite samples can be multiplexed for library preparation with the NEBNext DNA Ultra II library preparation kit.

65. Quantify the amplicon pool with Qubit 1x DNA HS. Up to 1 µg in 50 µl can be used as input for library preparation.

66. Prepare the end repair master mix, keep on ice.

Reagents for End Repair	Volume for 1 rxn
End Prep Enzyme Mix (green cap)	3 µl
End Prep Reaction Buffer (green cap)	7 µl
<i>Total</i>	<i>10 µl</i>

67. Add 10 µl end repair master mix to 50 µl purified amplicon DNA, mix briefly by pipetting and spin down. Run the sample with the following program on a thermal cycler with a heated lid.

Program setup	Temperature	Time
Step 1	20°C	30 min
Step 2	65°C	30 min
Step 3	4°C	Hold

68. Add the ligation reagents to the end prepped sample.

Reagents for Ligation	Volume to add
Ligation Master Mix (red cap)	30 µl
Ligation Enhancer (red cap)	1 µl
Adaptor (red cap)	2.5 µl

69. Mix by pipetting, quickly spin down and incubate 15 min at 20°C without a heated lid.

70. Add 3 µl USER, mix by pipetting, quickly spin down and incubate 15 min at 37°C with a heated lid.

CRITICAL STEP: The conditions for size selection of the adapter ligated library depend on the amplicon pool. In this case the amplicon range is from 300 bp up to 500 bp in size. Consult the Ultra II protocol for the size selection criteria suitable to your conditions.

71. Add 17.5 µl of AMPure XP beads to 96.5 µl of adapter ligated library and mix by pipetting about 10 times.

CRITICAL STEP: Prior to usage put AMPure XP beads at RT. To ensure a correct ratio is maintained mix the beads well on a vortexer shortly before adding them to the sample.

72. Incubate the sample for 5 min at RT.

73. Place the sample on a magnetic stand until the solution is clear. This will take about 5 min.

74. While keeping the tube on the magnetic stand transfer the supernatant to a new PCR tube.

75. Add another 17.5 µl of AMPure XP beads to the supernatant and mix by pipetting about 10 times.

76. Incubate the sample for 5 min at RT.

77. Place the sample on a magnetic stand until the solution is clear. This will take about 5 min.

78. While keeping the tube on the magnetic stand remove and discard the supernatant.

79. Wash the beads with 200 µl freshly prepared 80% EtOH, while keeping the sample on a magnetic stand.

CRITICAL STEP: Freshly prepared EtOH should not be older than 48 hrs as this will reduce the concentration of the washing solution and in effect the output of the sample.

80. Repeat washing step 79 once more.

81. After the second wash, remove the supernatant completely and lay the tubes on their side to air dry the beads. This will take approximately 1-3 min.

CRITICAL STEP: Air drying the beads is very critical. Avoid over drying the beads, as this would mean a substantial loss of material when rehydrating the beads.

82. Once the beads are no longer glossy add 17 µl 0.1X TE and resuspend the dried pellet by tapping the tube.

83. Incubate for 2 min.
84. Place the sample on a magnetic stand until the solution is clear. This will take about 2 min.
85. Transfer 15 µl of eluate to a new tube.
86. Add the following components to 15 µl of purified adapter-ligated library, up to a final volume of 50 µl.

CRITICAL STEP: Assign the indexing barcodes in such a way that optimal diversity is guaranteed. Consult the indexing list in the manufacturer's protocol.

Reagents for final PCR master mix	Volume to add
Q5 Master Mix (blue cap)	25 µl
Universal primer (blue cap)	5 µl
Indexing primer (blue cap)	5 µl

87. Mix by pipetting, spin down and run the sample with the following program on a thermal cycler with a heated lid.

Steps	# cycles	Time	Temperature
Initial denaturation	1	30 sec	98°C
Denaturation	3	10 sec	98°C
Annealing/extension		75 sec	65°C
Final extension	1	5 min	65°C
Hold		Indefinite	4°C

88. Clean up the final library by adding 45 µl of AMPure XP beads to 50 µl of the sample (0.9x ratio).
89. Follow steps 54-64 as described earlier in protocol to carry out the clean-up.
90. Elute the final library by adding 33 µl 0.1X TE and transfer 30 µl of eluate to a new tube.
91. Check the quality of the final library with a bioanalyzer DNA 1000 chip and the quantity with Qubit 1x DNA HS. A successful example is shown in figure 3b.
92. Run the sample on an Illumina sequencing platform, typically a Miseq 250 bp paired-end run gives sufficient results

? TROUBLESHOOTING

Computational analysis

Software installation

NOTE: All the steps related to the computational analysis of SMF data will depend on external software. Steps 93, 94, 97, 99 and 100 require bash command line usage. Steps 95, 96 and 98 involve code in R.

93. To install Trimmomatic, navigate to <http://www.usadellab.org/cms/?page=trimmomatic> and download the binary for the version of your choice. Uncompress Trimmomatic-0.xx.zip and move the resulting folder to a location of your choice. No further action is required for installation.
94. To install Picard, navigate to <https://broadinstitute.github.io/picard/>, uncompress the .zip file and move the .jar file to a location of your choice. No further action is required for installation.
95. Read alignment and downstream analysis will depend on the Biostrings-based genome data package for the species of interest. This can be installed as indicated in the relevant Bioconductor webpage and exemplified below for *Mus musculus*:

```
if (!requireNamespace("BiocManager", quietly = TRUE))
  install.packages("BiocManager")

BiocManager::install("BSgenome.Mmusculus.UCSC.mm10")
```

96. Analysis of single molecule footprinting data can be performed using the SingleMoleculeFootprinting R package, which can be installed as follows

```
remotes::install_github(
  repo = "https://github.com/Krebslabrep/SingleMoleculeFootprinting.git",
  ref = "main", build_vignettes = FALSE)
```

Raw data processing

97. Read trimming can be performed using Trimmomatic as exemplified by the following command line:

```
java -jar /installation_path/Trimmomatic-0.36/trimmomatic-0.36.jar \
  PE \
  -threads 5 \
  Sample_Name_R1.fq.gz Sample_Name_R2.fq.gz \
  Sample_Name_forward_paired.fq.gz \
  Sample_Name_forward_unpaired.fq.gz \
  Sample_Name_reverse_paired.fq.gz \
  Sample_Name_reverse_unpaired.fq.gz \
```

```
ILLUMINACLIP:/installation_path/Trimmomatic-0.36/adapters/TruSeq3-
PE.fa:2:30:10 \

LEADING:3 \

TRAILING:3 \

SLIDINGWINDOW:4:18
```

98. Read alignment can be performed using the following QuasR function:

```
library(QuasR)
library(BSgenome.Mmusculus.UCSC.mm10)
proj=qAlign(sampleFile=sampleSheet,

            genome="BSgenome.Mmusculus.UCSC.mm10",

            aligner="Rbowtie",

            projectName="ProjectName",

            paired="fr",

            bisulfite="undir",

            alignmentsDir="./",

            alignmentParameter="-e 70 -X 1000 -k 2 --best -strata",

            cacheDir=tempdir())
```

NOTE: The *sampleSheet* path to pass as *sampleFile* argument should point to a tab delimited file describing the location of the input fastq files. For more details, we refer to the *qAlign* documentation

<https://www.rdocumentation.org/packages/QuasR/versions/1.12.0/topics/qAlign>.

NOTE: Depending on the reference genome and sample coverage, this step can be quite demanding in terms of computational resources and execution times. This is particularly relevant in case the reference genome has not yet been bisulfite converted and indexed, in which case QuasR will perform these steps internally.

99. If applicable, technical replicates can be merged using the following Picard utility function:

```
java -Xmx4g -jar /installation_path/picard.jar MergeSamFiles \

    I=Replicate1.bam \

    I=Replicate2.bam \

    O=Merged.bam
```

100. Read deduplication (not recommended for amplicon experiments) can be performed as follows:

```
java -Xmx4g -jar /installation_path/picard.jar MarkDuplicates \

    INPUT=Merged.bam \

    OUTPUT=Merged_deduplicated.bam \
```

```
METRICS_FILE=Deduplication_stats.txt \
VALIDATION_STRINGENCY=LENIENT \
REMOVE_DUPLICATES=true \
TMP_DIR=TmpDir
```

Packages loading

NOTE: All the code below is executed in R. All the necessary tools, including QuasR, are imported with **SingleMoleculeFootprinting** except for the BSgenome of interest. We exemplify here the latter using the mouse genome.

101. Loading libraries

```
library(SingleMoleculeFootprinting)
library(BSgenome.Mmusculus.UCSC.mm10)
```

Library quality controls

102. To assess the quality of the sequencing libraries, we recommend inspecting canonical QC metrics that can be produced using the QuasR QC report as follows:

```
qQCReport(Qinput, pdfFilename=NULL, chunkSize=1e6L, useSampleNames=FALSE,
clobj=NULL)
```

For more details, we refer to the qQCReport documentation at <https://www.rdocumentation.org/packages/QuasR/versions/1.12.0/topics/qQCReport>.

103. The conversion rate measures the conversion of cytosines outside of methylated contexts and is expected to exceed 95%. It is generally sufficient to calculate it for a single chromosome as follows:

```
ConversionRate(sampleSheet, genome, chr)
```

104. If applicable, the efficiency of the bait capture process, which is expected to exceed 70%, can be verified as follows:

```
BaitCapture(sampleSheet, genome, baits)
```

SMF data analysis

105. Methylation calls at the single molecule level can be extracted for a region of interest as follows:

```
CallContextMethylation(sampleSheet, sample, genome, range, coverage, ConvRate.thr)
```

The function calls methylation events for the cytosines in the genomic context relevant for the experiment (single enzyme, double enzyme, etc.), filters reads based on conversion rate, collapses strands and filters cytosines for coverage.

The output consists of a list of two objects. The first is a GRanges object reporting the average methylation values for the covered cytosines in the region of interest. The second is a matrix reporting the binary methylation information at the single molecule level. This object can be used directly for plotting or for single read sorting as explained in the following steps.

CRITICAL: Note that the resulting matrix can require a large amount of memory. We therefore recommend to keep the region of interest, passed as the *range* parameter, short.

106. To sort single reads based on their occupancy patterns, the user can employ either of the following functions depending on whether the sorting is to be carried over a single transcription factor (TF) or multiple TFs.

`SortReadsBySingleTF(MethSM, TFBS)`

`SortReadsByTFCluster(MethSM, TFBSs)`

107. Three types of information can be visualized for a single locus: the average SMF signal, the single molecule stack and the state quantification. These can be obtained either individually or simultaneously using the following functions:

`PlotAvgSMF(MethGR, range, TFBSs)`

`PlotSM(MethSM, range, SortedReads)`

`StateQuantificationPlot(SortedReads)`

`PlotSingleSiteSMF(ContextMethylation, sample, range, SortedReads, TFBSs, saveAs)`

The parameter SortedReads for the last three functions can be set either to `NULL` to visualise unsorted reads, to `"HC"` to perform hierarchical clustering (useful to spot PCR duplicates) or to the output of either of the SortReads functions from the section above.

TROUBLESHOOTING

Step	Problem	Possible reason	Solution
34	Fragment size shifted towards higher molecular weight	DNA shearing is insufficient	Increase sonication by 20 to 40 sec. Otherwise repeat sonication. In case the issue remains, repeat the sonication with a new DNA extraction.
92/117	Footprinting is incomplete, methylation not saturated	Cell suspension not homogenous	Ensure proper cell dissociation, check the quality of the nuclear preparation and carefully verify the number of cells used.
		Nuclear extraction incomplete	Repeat nuclear extraction, check with microscopy. Specific cell types may require optimization.
		Methyl-transferase activity insufficient	Replace enzyme. Check activity by in-vitro methylation and digestion of genomic DNA with methyl-sensitive restriction enzymes such as HpaII for CpG methylation and BbvI for GpC methylation.
		Insufficient amounts of SAM in the reaction.	Use fresh SAM aliquots.
		Strong variation of footprinting between samples	Keep incubation times similar between samples. Also do not process more than 8 samples per round.

92/117	Low conversion rate	Conversion kit either too old or ineffective	Check and replace conversion kit. When using an already prepared conversion mix, do not keep it for longer than 1 week at -20°C.
92/117	High PCR duplication rate (>20%), low complexity library.	Too low input	Increase input material (>1 µg), pool several reactions if needed.
117	Low concentration of the final library	Failed capture	Repeat the hybridization, use fresh beads for capture. Check the baits; expiration date. Check the thermal cycler; ensure that the lid is set to 65°C to avoid sample evaporation.
117	Bait capture efficiency below 70%	No enrichment library	Repeat the hybridization, use fresh beads for capture. Check the baits; expiration date. Check the thermal cycler; ensure that the lid is set to 65°C to avoid sample evaporation.

TIMING

Day 1

Step 1-15, Footprinting the DNA: 2h30min

Day 2

Step 16-32, Isolating the DNA: 2h

Step 33-35 (B), Bisulfite conversion: 10min + 5h30min incubation

Day 3

Step 33-60 (A) Library preparation up to hybridization: 6h + 16h incubation

Step 33-64 (B) Bisulfite conversion, amplicon PCR and gel check: 2h50min

Day 4

Step 61-117 (A) Capture and final library preparation and sequencing: 8h30min

Step 65-92 (B) Library preparation and sequencing: 3h40min

ANTICIPATED RESULTS

Preparation of the libraries

A typical SMF reaction yields 0.7-1 μg of DNA. A minimum of 1 μg of DNA is required to prepare a genome-wide bisulfite library (with optional bait-capture). We recommend performing three SMF reactions in parallel yielding 2-3 μg of input DNA for the library preparation (corresponding to 0.75×10^6 mouse/human cells and 7.5×10^6 Drosophila cells). A successful library preparation yields 60-120 ng of DNA. $\sim 1 \mu\text{g}$ of DNA is recommended to perform an amplicon bisulfite library over 96 targets. The amount of DNA can be scaled down when targeting fewer loci. However, reducing the input DNA in individual reactions can lead to a reduction of library complexity and an increase of the proportion of identical molecules in the resulting datasets. Decreased complexity can be detected by an increase of the duplication rates above 20% in genome-wide experiments. This can be circumvented by increasing the DNA used as an input to the reaction.

Sequencing depth requirements

Accurate quantification of a TF binding at single molecule resolutions requires to generate sufficient sequencing reads spanning all the bins surrounding individual TFBS (Figure 5a). A minimal coverage of 40 reads is recommended. This is sufficient to reproducibly quantify binding frequencies above 20%, where 8 reads out of 40 reads would have a TF footprint. Lower binding frequencies will have lower counts of bound molecules, and accurate quantification of the binding frequency requires to increase the total number of molecules sequenced at the locus. Moreover, when quantifying the binding of multiple TFs in a cluster, the classification algorithm will only consider the reads spanning all the studied binding sites (Figure 5b). Therefore, it is important to adjust coverage as a function of the number of loci considered, the binding frequency of the TFs analysed and the distance between the TFBS in a cluster. We recommend multiplexing a maximum of two SMF samples on a NextSeq 500, producing $\sim 200 \times 10^6$ read pairs of length 150 bp, sequencing a total of 6×10^{10} nucleotides. Out of these about $\sim 40\%$ will not map, $\sim 20\%$ will be removed as PCR duplicates, leaving $\sim 2.88 \times 10^{10}$ analysable nucleotides. This represents a theoretical coverage of ~ 144 times the drosophila genome and ~ 360 times the 80Mb of the mouse genome captured by the baits. However, reads are not evenly distributed in the genome, particularly when a capture step is applied, as pulldown efficiency varies between baits. Moreover, only a fraction of the reads covering a locus will span all the bins used to analyse the binding of one or more TFs (Figure 5). In practice, this amount of sequencing leads to 50-100 reads usable to quantify TF binding.

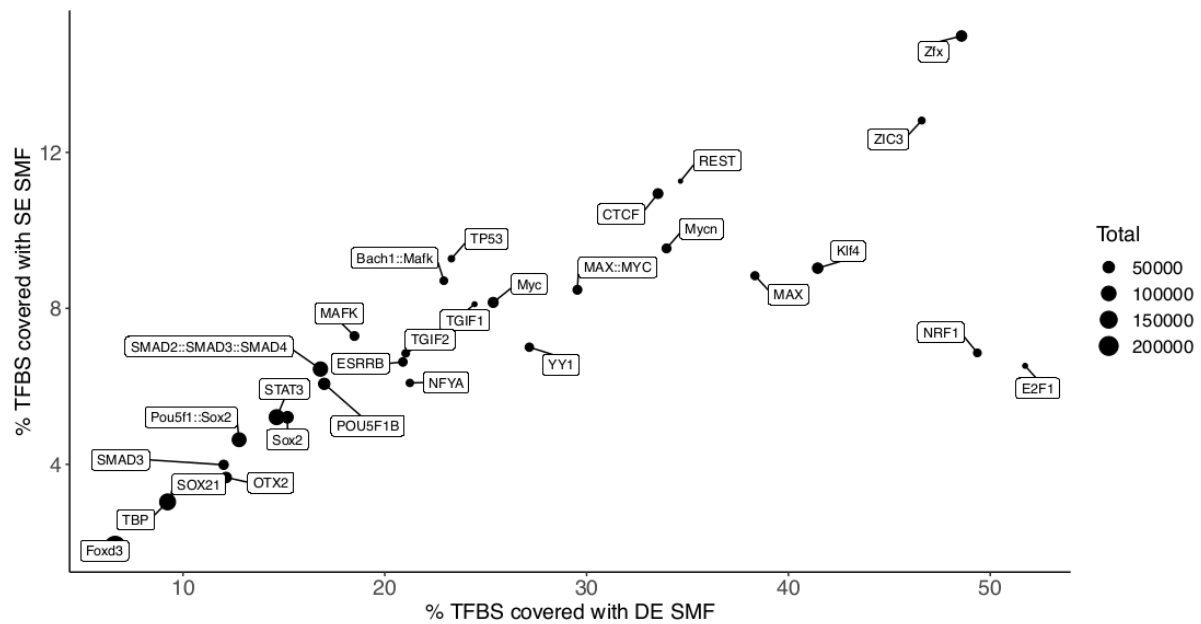


Figure 7. Number of TFBSs that can be studied by SMF. Classification of the single molecules at a TFBS requires the presence of informative cytosines in each of the classification bins. The scatterplot shows the percentages of TFBSs that can be analysed when performing SMF with the GpC methyltransferase McviPI (single enzyme - SE, y axis) or in combination with the CpG methyltransferase SssI (double enzyme - DE, x axis). The percentages are calculated with respect to the total number of TFBSs (dot size) mapped to the mouse genome using JASPAR PWMs with ChIP-seq evidence. For TFs such as NRF1, E2F1 and Klf4 there is quite a clear advantage in performing DE SMF as compared to SE SMF.

Number of TF binding sites analysed

SMF allows single molecule quantification of the binding of a TF at a fraction of its binding sites (Figure 7). The ability to analyse a given TFBS is defined by the sequence composition at its flanks. The classification of reads in various states requires to measure methylation in each of the bins used for sorting (Figure 5). Each bin should contain informative cytosines that are in the GpC context when performing the experiment with McviPI, and in combination with CpGs when performing the treatment with SssI. Combining the two methyltransferases significantly extends the number of TFBS that can be analysed (Figure 7). The extent of the increase varies depending on the identity of the TF. For instance, there is only a moderate increase in the number of REST sites quantified in a double enzyme experiment (from ~200 to ~600 sites) while between seven and eight times more NRF1 binding sites can be analysed (from ~900 to ~6700) (Figure 7). The number of analysable binding sites is in the range of thousands for most TFs. This is sufficient to determine their global binding properties, and visualise their behaviour at example loci. However, it is important to check the dinucleotide composition of target loci before performing an SMF experiment.

DATA AVAILABILITY

A test dataset is available from GitHub

(<https://github.com/Krebslabrep/SingleMoleculeFootprinting>).

CODE AVAILABILITY

All codes are available from GitHub

(<https://github.com/Krebslabrep/SingleMoleculeFootprinting>).

REFERENCES

1. Raha, D., Hong, M. & Snyder, M. ChIP-Seq: A Method for Global Identification of Regulatory Elements in the Genome. *Curr. Protoc. Mol. Biol.* **91**, (2010).
2. Skene, P. J. & Henikoff, S. An efficient targeted nuclease strategy for high-resolution mapping of DNA binding sites. *eLife* **6**, e21856 (2017).
3. Song, L. & Crawford, G. E. DNase-seq: A High-Resolution Technique for Mapping Active Gene Regulatory Elements across the Genome from Mammalian Cells. *Cold Spring Harb. Protoc.* **2010**, pdb.prot5384-pdb.prot5384 (2010).
4. Buenrostro, J. D., Giresi, P. G., Zaba, L. C., Chang, H. Y. & Greenleaf, W. J. Transposition of native chromatin for fast and sensitive epigenomic profiling of open chromatin, DNA-binding proteins and nucleosome position. *Nat. Methods* **10**, 1213–1218 (2013).
5. Reiter, F., Wienerroither, S. & Stark, A. Combinatorial function of transcription factors and cofactors. *Curr. Opin. Genet. Dev.* **43**, 73–81 (2017).
6. Morgunova, E. & Taipale, J. Structural perspective of cooperative transcription factor binding. *Curr. Opin. Struct. Biol.* **47**, 1–8 (2017).
7. Ibarra, I. L. *et al.* Mechanistic insights into transcription factor cooperativity and its impact on protein-phenotype interactions. *Nat. Commun.* **11**, 124 (2020).
8. Sönmezer, C. *et al.* Molecular Co-occupancy Identifies Transcription Factor Binding Cooperativity In Vivo. *Mol. Cell* **81**, 255-267.e6 (2021).
9. Kelly, T. K. *et al.* Genome-wide mapping of nucleosome positioning and DNA methylation within individual DNA molecules. *Genome Res.* **22**, 2497–2506 (2012).
10. Krebs, A. R. *et al.* Genome-wide Single-Molecule Footprinting Reveals High RNA Polymerase II Turnover at Paused Promoters. *Mol. Cell* **67**, 411-422.e4 (2017).
11. Nabils, N. H. *et al.* Multiplex mapping of chromatin accessibility and DNA methylation within targeted single molecules identifies epigenetic heterogeneity in neural stem cells and glioblastoma. *Genome Res.* **24**, 329–339 (2014).
12. Stergachis, A. B., Debo, B. M., Haugen, E., Churchman, L. S. & Stamatoyannopoulos, J. A. Single-molecule regulatory architectures captured by chromatin fiber sequencing. *Science* **368**, 1449–1454 (2020).
13. Lee, I. *et al.* Simultaneous profiling of chromatin accessibility and methylation on human cell lines with nanopore sequencing. *Nat. Methods* **17**, 1191–1199 (2020).
14. Abdulhay, N. J. *et al.* Massively multiplex single-molecule oligonucleosome footprinting. *eLife* **9**, e59404 (2020).
15. Shipony, Z. *et al.* Long-range single-molecule mapping of chromatin accessibility in eukaryotes. *Nat. Methods* **17**, 319–327 (2020).

16. Gaidatzis, D., Lerch, A., Hahne, F. & Stadler, M. B. QuasR: quantification and annotation of short reads in R. *Bioinformatics* **31**, 1130–1132 (2015).
17. Langmead, B., Trapnell, C., Pop, M. & Salzberg, S. L. Ultrafast and memory-efficient alignment of short DNA sequences to the human genome. *Genome Biol.* **10**, R25 (2009).
18. <http://broadinstitute.github.io/picard/>.
19. Fornes, O. *et al.* JASPAR 2020: update of the open-access database of transcription factor binding profiles. *Nucleic Acids Res.* gkz1001 (2019) doi:10.1093/nar/gkz1001.
20. Puig, R. R., Boddie, P., Khan, A., Castro-Mondragon, J. A. & Mathelier, A. *UniBind: maps of high-confidence direct TF-DNA interactions across nine species.* <http://biorxiv.org/lookup/doi/10.1101/2020.11.17.384578> (2020) doi:10.1101/2020.11.17.384578.
21. Minnoye, L. *et al.* Chromatin accessibility profiling methods. *Nat. Rev. Methods Primer* **1**, 10 (2021).
22. Buenrostro, J. D. *et al.* Single-cell chromatin accessibility reveals principles of regulatory variation. *Nature* **523**, 486–490 (2015).
23. Cusanovich, D. A. *et al.* Multiplex single-cell profiling of chromatin accessibility by combinatorial cellular indexing. *Science* **348**, 910–914 (2015).
24. Levo, M. *et al.* Systematic Investigation of Transcription Factor Activity in the Context of Chromatin Using Massively Parallel Binding and Expression Assays. *Mol. Cell* **65**, 604–617.e6 (2017).
25. Oberbeckmann, E. *et al.* Absolute nucleosome occupancy map for the *Saccharomyces cerevisiae* genome. *Genome Res.* **29**, 1996–2009 (2019).
26. Stadler, M. B. *et al.* DNA-binding factors shape the mouse methylome at distal regulatory regions. *Nature* **480**, 490–495 (2011).
27. Domcke, S. *et al.* Competition between DNA methylation and transcription factors determines binding of NRF1. *Nature* **528**, 575–579 (2015).

CONTRIBUTIONS

A.R.K designed the study. R.K., G.B. and A.R.K wrote the manuscript. R.K performed the experiments. G.B. developed the package for data analysis with support from M.L.S. A.R.K supervised conduction of the experiments and the data analysis. All authors discussed the results and commented on the manuscript.

ACKNOWLEDGMENTS

The authors are grateful to the members of the Krebs laboratory for helpful discussions, comments on the manuscript and feedback during the development of the R package. The authors would like to acknowledge Can Sönmezer for sharing data and Elisa Kreibich for sharing amplicon QC gels. The authors are thankful to Laura Villacorta, Vladimir Benes and the members of the Genomics Core facility for sequencing the libraries and technical assistance. The salary of G.B. is supported by the Deutsche Forschungsgemeinschaft (KR 5247/1-1). The authors thank Wolfgang Huber for supporting the development of the R package. The authors would like to thank Charles Girardot and the Genome Biology Computational Support. Research in the laboratory of A.R.K is supported by core funding of the European Molecular Biology Laboratory, Deutsche Forschungsgemeinschaft (KR 5247/1-

1). M.L.S. is funded by The German Network for Bioinformatics Infrastructure (de.NBI) Förderkennzeichen Nr. 031A537B.

COMPETING INTERESTS

The authors declare no competing interests.

ADDITIONAL INFORMATION

Correspondence and requests for materials should be addressed to A.R.K.
arnaud.krebs@embl.de

RELATED LINKS

Key reference using this protocol

Sönmezer C, Kleinendorst R, Imanci D, Barzaghi G, Villacorta L, Schübeler D, Benes V, Molina N, Krebs AR. Molecular Co-occupancy Identifies Transcription Factor Binding Cooperativity In Vivo. *Mol Cell*. 2021 Jan;81(2) 255-267.e6. doi:10.1016/j.molcel.2020.11.015. PMID: 33290745.

Krebs AR, Imanci D, Hoerner L, Gaidatzis D, Burger L, Schübeler D. Genome-wide Single-Molecule Footprinting Reveals High RNA Polymerase II Turnover at Paused Promoters. *Mol Cell*. 2017 Aug;67(3) 411-422.e4. doi:10.1016/j.molcel.2017.06.027. PMID: 28735898; PMCID: PMC5548954.